AHMED OUMAR EL-SHANGITI

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RESEARCH SUMMARY

Mechanistic interpretability of LLMs and VLMs, focusing on numerical reasoning and counting. Publications at ACL, NAACL, and EMNLP (incl. **ACL 2025 Best Resource Paper**); interpretability work **cited by Anthropic, Harvard, and UC Berkeley**; invited talks at **Imperial College London** and **Grammarly**, among others.

EDUCATION

MSc in Natural Language Processing, MBZUAI 2025 – Present
Relevant coursework: Advanced NLP, Multimodal Modeling, Interpretability of LLMs, Agentic AI.

B.Sc. in Mathematics and Computer Science, Cadi Ayyad University 2021

EXPERIENCE

Co-Founder & CEO Dec 2025 – Present
DataBayt.AI *Abu Dhabi, UAE*

- Leading development of open-source software and datasets for data-centric AI.
- databayt.ai | github.com/DataBayt-AI

Guest Researcher Jun 2025 – Aug 2025
UKP Lab, Technical University of Darmstadt *Darmstadt, Germany*

- Conducted research on the interpretability of Vision-Language Models (VLMs).

NLP Research Fellow Aug 2024 – Aug 2025
RIKEN Center for Advanced Intelligence Project (AIP) *Japan*

- Research on the interpretability of Large Language Models.

NLP Research Associate Mar 2023 – Aug 2025
MBZUAI *Abu Dhabi, UAE*

- Published papers at top-tier venues (ACL, EMNLP, NAACL), including an ACL 2025 Best Resource Paper.
- Worked on generative AI for low-resource languages, machine translation, and dataset design.
- Investigated reasoning mechanisms and interpretability in LLMs.

NLP Research Assistant Intern Oct 2022 – Feb 2023
Qatar Computing Research Institute (QCRI) *Remote*

- Developed state-of-the-art propaganda detection systems for Arabic and multilingual settings.
- Ranked 2nd in SemEval 2023 Task 3.

NLP Engineer Intern Feb 2022 – Jun 2022
Walmart Global Tech *Remote*

- Optimized production-scale language models for product type prediction.
- Reduced model size by 40–60% while maintaining performance.

SELECTED PUBLICATIONS

The Count Is There, but Misaligned: Understanding and Correcting Counting Failures in VLMs

- **El-Shangiti** et al., 2026 (under review)
- *Cited by UC Berkeley: citing paper*

The Geometry of Numerical Reasoning: Language Models Compare Numeric Properties in Linear Subspaces

- **El-Shangiti** et al., NAACL 2025 (oral)
- *Cited by Harvard: citing paper*

Number Representations in LLMs: A Computational Parallel to Human Perception

- H. V. AlquBoj et al., 2025 (arXiv)
- *Cited by Anthropic: citing paper*

Palm: A Culturally Inclusive and Linguistically Diverse Dataset for Arabic LLMs

- Alwajih et al., ACL 2025 — **Best Resource Paper Award**

Arabic Automatic Story Generation with Large Language Models

- **El-Shangiti** et al., ArabicNLP Conference 2024 (oral)

Casablanca: Data and Models for Multidialectal Arabic Speech Recognition

- Talafha et al., EMNLP 2024

Dolphin: A Challenging and Diverse Benchmark for Arabic NLG

- Nagoudi et al., Findings of EMNLP 2023

SKILLS, TALKS, AND AWARDS

Skills	Python, PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, NLTK, Git/GitHub
Invited Talks	Imperial College London; Grammarly; among others
Workshops	Generative AI in the Era of LLMs (I2COMSAPP, Apr 2024); UBC NLP Workshop
Awards	**Best Resource Paper Award**, ACL 2025; Facebook Secure and Private AI Scholarship, 2019; 1st place, Arabic Misogyny Identification Shared Task; 2nd place, SemEval 2023 Task 3
Languages	Arabic, English, French, German, Japanese, Urdu